

APPLETHORPE, Australia

WMO: 945530

Lat: 28.6217S

Lon: 151.9533E

Elev: 873

StdP: 91.27

Time Zone: 10.00 (E10)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -2.3	(c) -0.9	(d) -8.2	(e) 2.1	(f) 15.1	(g) -6.0	(h) 2.5	(i) 13.6	(j) 8.0	(k) 8.5	(l) 7.2	(m) 8.4	(n) 0.3	(o) 50	(p) 0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 10.5	(c) 31.1	(d) 18.2	(e) 29.4	(f) 17.9	(g) 27.8	(h) 17.4	(i) 20.9	(j) 26.0	(k) 20.2	(l) 24.9	(m) 19.6	(n) 24.0	(o) 3.4	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.6	(b) 16.0	(c) 22.3	(d) 19.0	(e) 15.3	(f) 21.7	(g) 18.3	(h) 14.7	(i) 21.1	(j) 64.9	(k) 25.9	(l) 62.2	(m) 25.0	(n) 60.0	(o) 24.0	(p) 27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.0	6.1	5.4	-5.0	34.0	1.2	1.9	-5.9	35.4	-6.5	36.5	-7.2	37.5	-8.0	38.9
			-2.4	22.7	1.5	1.9	-3.5	24.1	-4.4	25.2	-5.2	26.3	-6.3	27.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	21.0	20.2	18.7	15.3	11.4	9.0	8.1	9.2	12.8	15.6	18.1	19.8		
	DBStd	5.20	2.45	2.37	2.07	2.31	2.57	2.59	2.42	2.59	2.96	2.94	2.94	2.73		
	HDD10.0	186	0	0	0	1	15	49	68	46	7	1	0	0		
	HDD18.3	1532	5	6	20	93	214	280	316	284	168	93	38	15		
	CDD10.0	1980	340	285	271	161	59	19	10	21	91	176	243	304		
	CDD18.3	283	87	58	32	3	0	0	0	1	2	9	31	61		
	CDH23.3	2217	641	388	198	20	2	0	0	4	38	127	306	494		
	CDH26.7	591	202	111	37	1	0	0	0	1	4	20	76	139		
(14) Wind	WSAvg	2.4	2.7	2.6	2.6	2.2	2.0	2.3	2.3	2.4	2.5	2.5	2.5	2.6		
(15) Precipitation	PrecAvg	784	100	84	66	44	53	39	51	44	45	72	84	98		
	PrecMax	1092	248	255	228	242	232	104	144	127	117	238	255	214		
	PrecMin	527	21	4	0	1	3	3	1	0	0	16	22	18		
	PrecStd	156	62	50	50	49	48	25	39	31	30	44	48	50		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	32.9	30.1	25.7	22.0	18.3	18.6	22.2	26.9	29.1	31.2	32.4		
		MCWB	18.7	19.3	17.8	15.7	13.6	12.2	10.9	10.9	14.2	14.4	17.6	17.9		
	2%	DB	30.8	29.9	27.4	23.5	20.0	16.8	16.6	19.2	24.1	26.5	28.9	30.1		
		MCWB	18.6	19.2	17.8	15.3	12.9	11.3	10.2	10.1	13.1	14.4	16.6	17.7		
	5%	DB	29.0	27.6	25.5	22.0	18.5	15.6	15.3	17.2	22.1	24.6	26.8	28.0		
		MCWB	18.2	18.5	17.6	14.9	12.2	10.8	9.9	9.5	12.4	14.1	15.9	17.3		
	10%	DB	27.0	25.7	23.8	20.6	17.1	14.3	14.0	15.6	19.8	22.5	24.7	26.2		
		MCWB	18.3	18.4	17.2	14.6	11.7	10.4	9.2	8.9	11.8	13.7	15.5	17.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.8	22.1	20.5	18.1	16.1	14.5	14.9	13.8	16.7	18.5	20.2	21.3		
		MCDB	27.1	27.7	24.8	21.9	18.7	16.2	16.5	18.9	20.6	22.8	26.7	27.1		
	2%	WB	20.9	20.9	19.7	17.1	14.8	13.2	13.1	12.4	14.8	17.0	18.8	20.1		
		MCDB	25.9	25.7	23.5	20.6	17.5	15.0	14.7	15.5	20.0	21.4	24.1	25.2		
	5%	WB	20.2	20.0	19.0	16.3	13.8	12.3	11.5	11.4	13.8	16.0	17.9	19.2		
		MCDB	24.8	24.4	22.5	19.6	16.5	14.0	13.4	14.8	18.7	20.8	22.6	23.9		
	10%	WB	19.6	19.2	18.3	15.5	13.0	11.4	10.4	10.5	13.0	15.1	17.1	18.5		
		MCDB	24.0	23.3	21.6	18.7	15.4	13.2	12.6	14.0	17.8	19.7	21.9	23.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	9.6	9.7	10.4	11.5	10.2	11.4	12.9	13.0	12.1	11.4	10.8		
		MCDBR	14.5	13.8	13.1	12.7	13.7	12.6	14.0	15.7	16.7	15.9	15.0	14.5		
	5% WB	MCWBR	4.2	4.3	4.9	5.4	6.9	6.6	7.1	6.9	6.9	5.7	4.7	4.4		
		MCWBR	10.5	10.5	9.9	10.0	9.4	8.3	8.8	10.2	12.4	11.4	10.9	10.8		
(40) Clear-Sky Solar Irradiance	taub		0.340	0.329	0.311	0.295	0.275	0.270	0.264	0.275	0.308	0.329	0.342	0.343		
		taud	2.571	2.623	2.674	2.689	2.697	2.711	2.716	2.658	2.540	2.511	2.519	2.548		
	Ebn at Noon		1001	995	984	955	932	916	937	963	970	984	993	1001		
		Edh at Noon	107	99	89	79	71	66	68	80	100	109	112	110		
(44) All-Sky Solar Radiation	RadAvg		6.82	6.18	5.31	4.57	3.72	3.08	3.49	4.45	5.50	6.21	6.69	6.80		
	RadStd		0.54	0.54	0.33	0.20	0.23	0.23	0.26	0.32	0.45	0.58	0.65	0.65		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-60	+172	+109

Nomenclature: See separate page